

Question 1 : Define Power BI and What are the key components of the Power BI ecosystem? Briefly explain:

- **Power BI Desktop**
- **Power BI Service**
- **Power BI Mobile**
- **Power BI Gateway**

ANSWER: Power BI is a business analytics and data visualization tool by Microsoft that helps users connect to data sources, transform data, create interactive reports and dashboards, and share insights across an organization to support data-driven decision-making.

Key Components of the Power BI Ecosystem

1. Power BI Desktop

- **A Windows-based application used to connect, transform, and model data.**
 - **Allows users to create reports and visualizations using drag-and-drop features.**
 - **Reports created in Power BI Desktop can be published to the Power BI Service.**
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2. Power BI Service

- **A cloud-based (SaaS) platform accessed through a web browser.**
- **Used to publish, share, collaborate, and manage reports and dashboards.**

- **Enables features like scheduled data refresh, dashboards, data sharing, and security.**
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3. Power BI Mobile

- **Mobile applications available for Android, iOS, and Windows devices.**
 - **Allows users to view and interact with reports and dashboards on the go.**
 - **Provides real-time alerts and notifications for important data changes.**
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4. Power BI Gateway

- **Acts as a bridge between on-premises data sources and Power BI Service.**
 - **Enables secure data refresh and live queries without moving data to the cloud.**
 - **Commonly used for SQL Server, Oracle, and other on-premises databases.**
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Summary

Together, these components allow users to build reports (Desktop), publish and collaborate (Service), access insights anywhere (Mobile), and securely connect on-premises data (Gateway).

Question 2 : Compare the following Power BI visuals: • Pie Chart vs Donut Chart

• Bar Chart vs Column Chart When would you prefer one over the other?

Give one example for each pair.

ANSWER: Pie Chart vs Donut Chart

Pie Chart

- Displays data as slices of a circle showing part-to-whole relationships.
- Simple and easy to understand when there are few categories.

Donut Chart

- Similar to a pie chart but with a hole in the center.
- The center space can be used to display total values or key labels, improving readability.

When to prefer which?

- **Pie Chart:** When you want a simple, quick comparison of proportions.
- **Donut Chart:** When you want a cleaner design and to highlight totals or KPIs.

Example

- **Showing market share of 3–4 companies:**
 - **Pie Chart:** Market share percentage
 - **Donut Chart:** Market share with total sales shown in the center
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Bar Chart vs Column Chart

Bar Chart

- **Displays data using horizontal bars.**
- **Best for long category names or when comparing many categories.**

Column Chart

- **Displays data using vertical columns.**
- **Best for time-based data or when categories follow a natural order.**

When to prefer which?

- **Bar Chart:** When labels are long or comparisons are across many items.
- **Column Chart:** When showing trends over time or ordered categories.

Example

- **Comparing sales by product:**
 - **Bar Chart:** Sales by product names with long labels
 - **Column Chart:** Monthly sales trend over a year
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Summary

- **Pie vs Donut** → Same purpose, donut offers better readability and space usage.
 - **Bar vs Column** → Orientation matters: bar for readability, column for trends.
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Question 3 : Explain the significance of:

- **Star schema vs Snowflake schema**
- **Primary key vs Foreign key in relationships (Power BI)**

Why is cardinality important?

ANSWER: Star Schema vs Snowflake Schema

Star Schema

- **Consists of a central fact table connected directly to dimension tables.**
- **Simple structure with fewer joins, making it faster and easier to understand.**
- **Preferred in Power BI for better performance and simpler DAX.**

Snowflake Schema

- **Dimension tables are normalized into multiple related tables.**
- **More complex with more joins, which can impact performance.**
- **Used when there is a need to reduce data redundancy.**

Significance

- **Star schema improves query performance, report simplicity, and model clarity.**
- **Snowflake schema is more storage-efficient but less optimal for Power BI models.**

Primary Key vs Foreign Key (Power BI Relationships)

Primary Key

- A column with unique values in a table.
- Identifies each record uniquely (usually in a dimension table).

Foreign Key

- A column that references the primary key in another table.
- Used to link fact tables to dimension tables.

Significance

- Primary keys ensure uniqueness, while foreign keys create relationships.
 - Correct key usage enables accurate filtering and aggregation in reports.
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Why Is Cardinality Important?

Cardinality defines the nature of the relationship between tables:

- One-to-One (1:1)
- One-to-Many (1:*) (*most common in Power BI*)
- Many-to-Many (:)

Importance

- **Determines how filters flow between tables.**
 - **Affects calculation accuracy, model performance, and DAX behavior.**
 - **Incorrect cardinality can cause duplicate values, wrong totals, or incorrect insights.**
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Summary

- **Star schema is preferred over snowflake in Power BI for performance and simplicity.**
 - **Primary keys and foreign keys define clear and reliable relationships.**
 - **Cardinality is crucial for correct data modeling and accurate analytics.**
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Question 4 : Differentiate between:

- **Calculated column vs Measure Also, define Row context and Filter context with simple examples.**

ANSWER: Calculated Column vs Measure

Calculated Column

- Calculated row by row and stored in the data model.
- Evaluated during data refresh.
- Uses row context.
- Increases model size.
- Best for static calculations (e.g., categorization, flags).

Example:

Profit = Sales[Revenue] - Sales[Cost] (calculated for each row)

Measure

- Calculated on the fly when a report is viewed.
- Evaluated based on filters and visuals.
- Uses filter context.
- Does not increase model size.
- Best for dynamic aggregations (e.g., totals, averages).

Example:

Total Sales = SUM(Sales[Revenue])

Row Context

Definition:

Row context means Power BI evaluates a calculation one row at a time.

Example:

In a calculated column:

Discount Amount = Sales[SalesAmount] * 0.1

Each row's **SalesAmount** is multiplied by 10%.

Filter Context

Definition:

Filter context is the set of filters applied to a calculation from:

- Visuals (charts, tables)
- Slicers
- Report/page filters

Example:

If a measure **Total Sales** is shown:

- Without filters → shows overall sales
 - With a slicer set to Year = 2024 → shows sales for 2024 only
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Summary Table

Feature	Calculated Column	Measure
Evaluation time	Data refresh	Query time
Storage	Stored in model	Not stored
Context used	Row context	Filter context
Best use	Row-level logic	Aggregations & KPIs

Key Takeaway

- **Use calculated columns for row-level logic.**
 - **Use measures for dynamic, filter-based calculations.**
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Question 5: What is the difference between a report and a dashboard in Power BI?

ANSWER: In Power BI, a report and a dashboard serve different purposes and are used at different levels of analysis.

A report is a detailed and interactive collection of visuals created from a single dataset. It can contain multiple pages, allowing users to explore data deeply using filters, slicers, drill-down, and drill-through features. Reports are mainly used for analysis, helping users understand trends, patterns, and detailed breakdowns of data. Reports are typically created in Power BI Desktop and then published to the Power BI Service.

A dashboard, on the other hand, is a single-page overview designed to show the most important insights at a glance. It is made by pinning visuals from one or more reports and datasets into one place. Dashboards have limited interactivity compared to reports and are mainly used for monitoring performance and KPIs. Dashboards are created only in the Power BI Service and are commonly used by managers and executives.

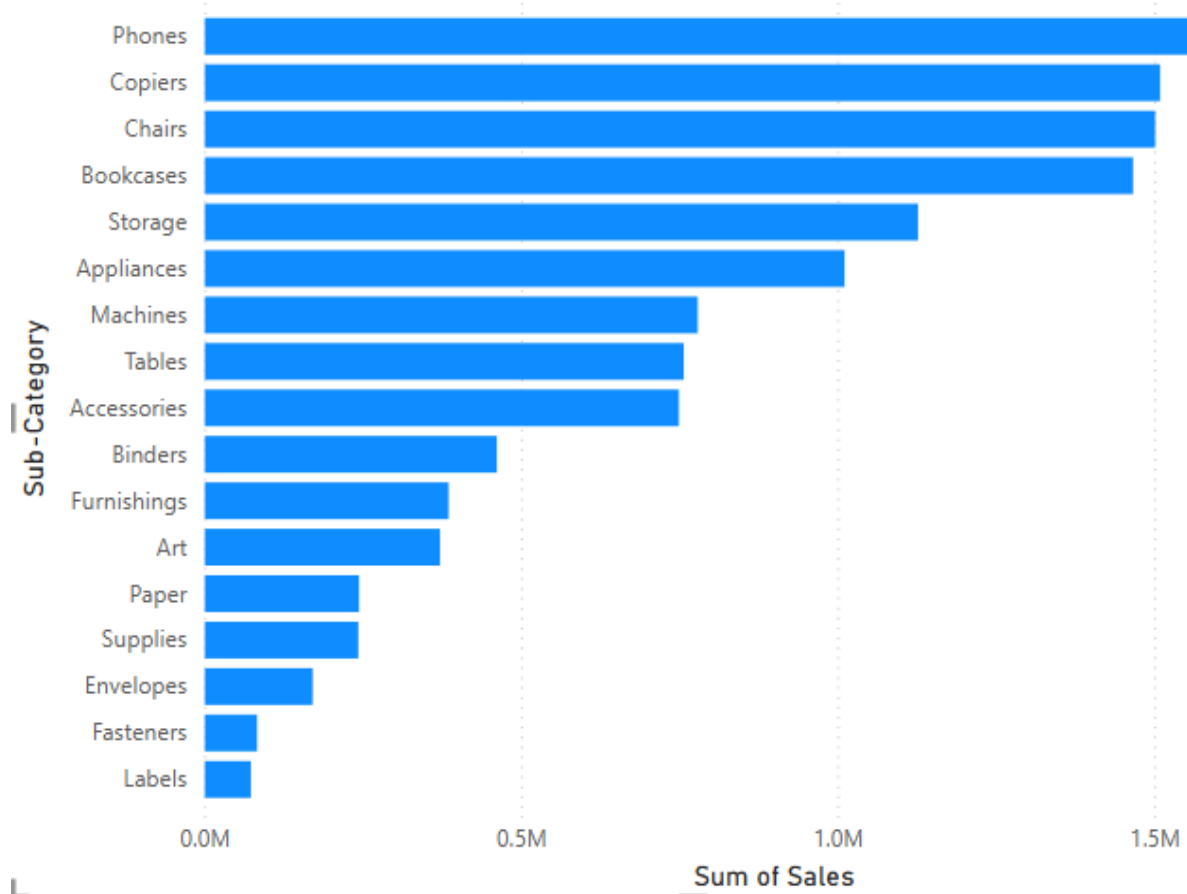
In simple terms, reports are for detailed analysis, while dashboards are for quick summaries and high-level decision-making.

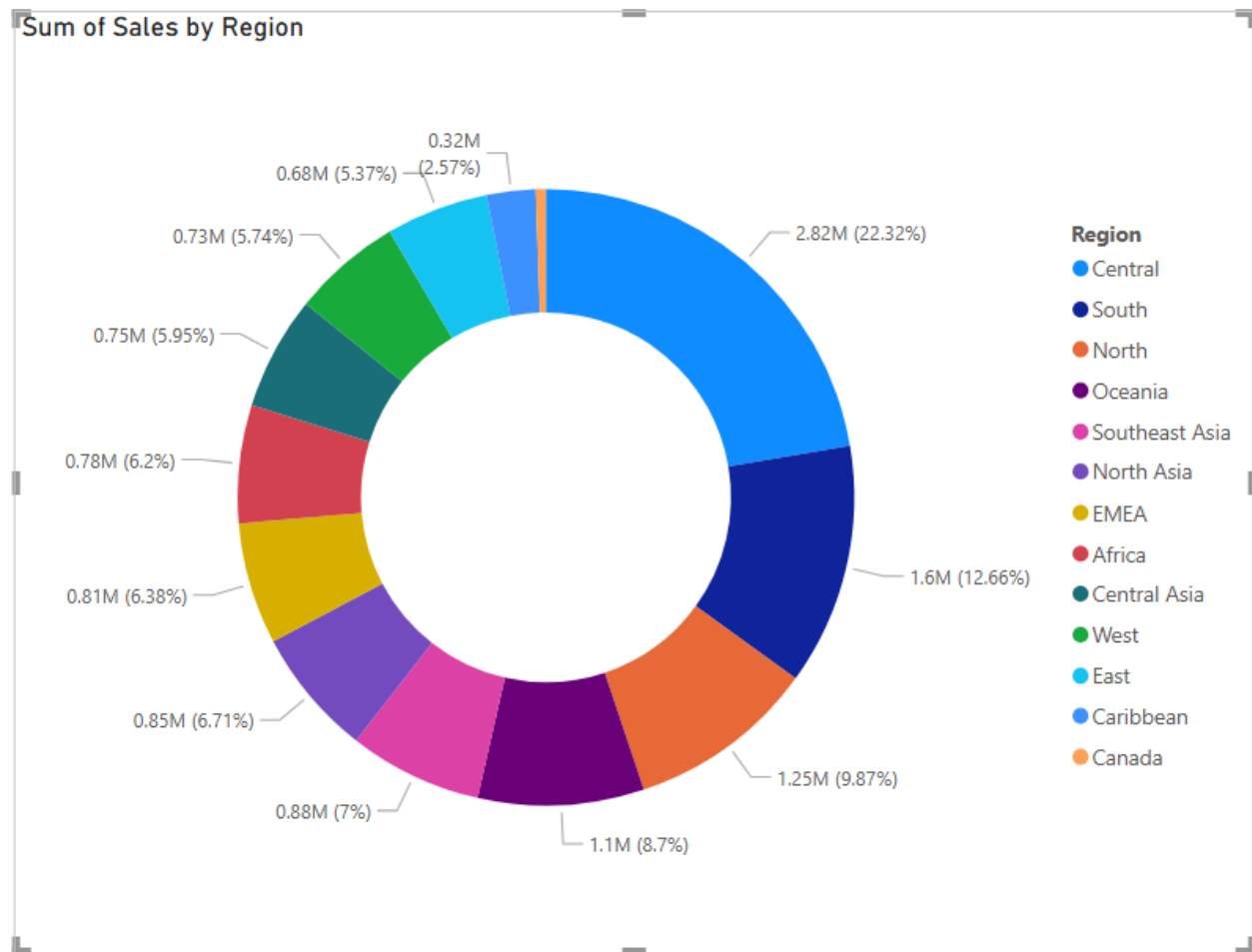
Question 6 : Using the Sample Superstore dataset:

- Create a Clustered Bar Chart to display Total Sales by Sub-Category
- Create a Donut Chart for Sales % by Region Provide screenshots of both visuals.

ANSWER:

Sum of Sales by Sub-Category

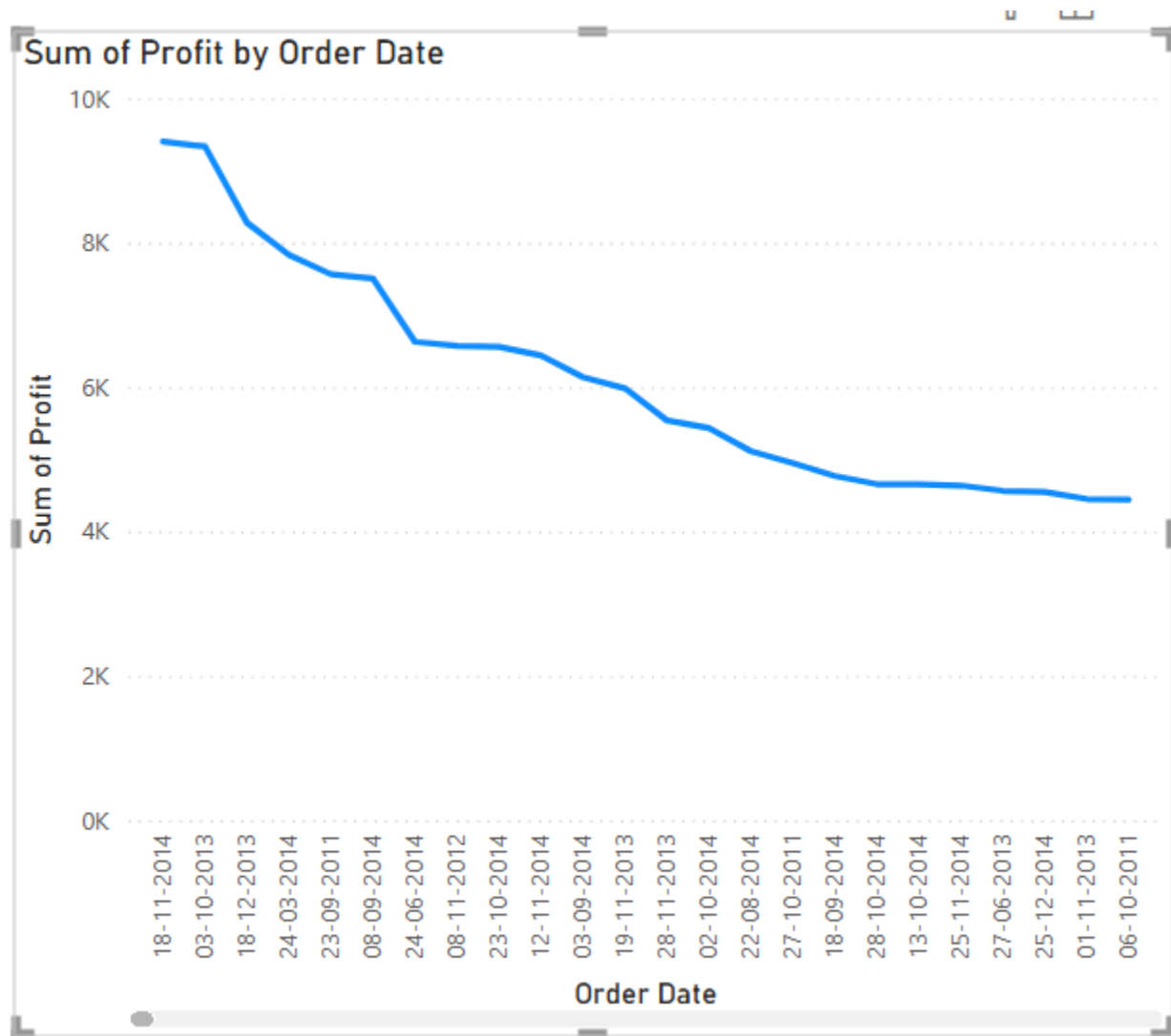




Question 7 : Write and apply the following measures: • Total Profit = SUM([Profit])

• Average Discount = AVERAGE([Discount]) Display both in a KPI Card, and use a Line Chart to show profit trend over months. Add visuals and DAX formulas.

ANSWER:

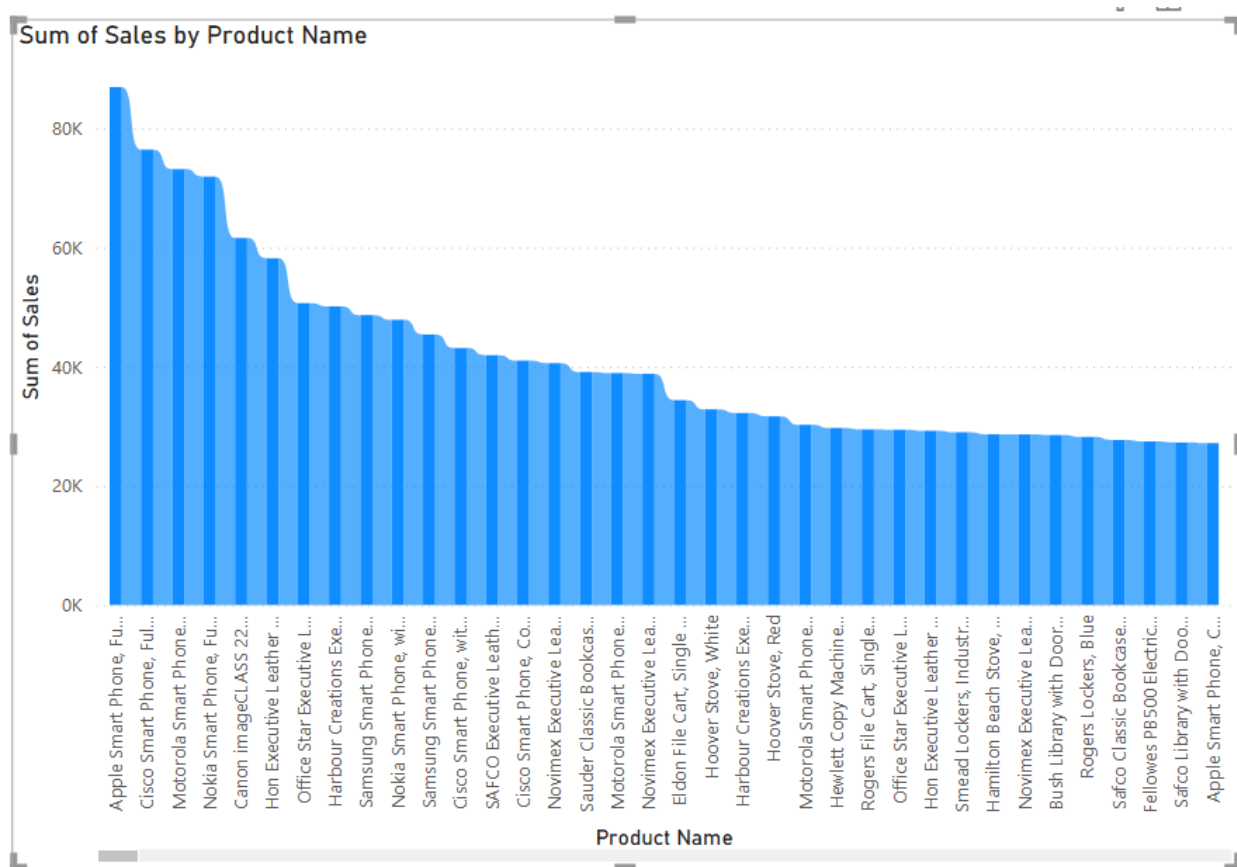


Question 8 : Implement a DAX measure that calculates the percentage of total sales by product category.

Product_category	Sales_Amount
Electronics	5000
Clothing	3000
Home Appliances	7000
Books	2000

Tables & Chairs	8000
Toy	1500
Sports Equipment	1200
Office Supplies	1000
Beauty Products	4400
Garden Supplies	1000
Jewelry	1800
Automotive	2600

ANSWER:

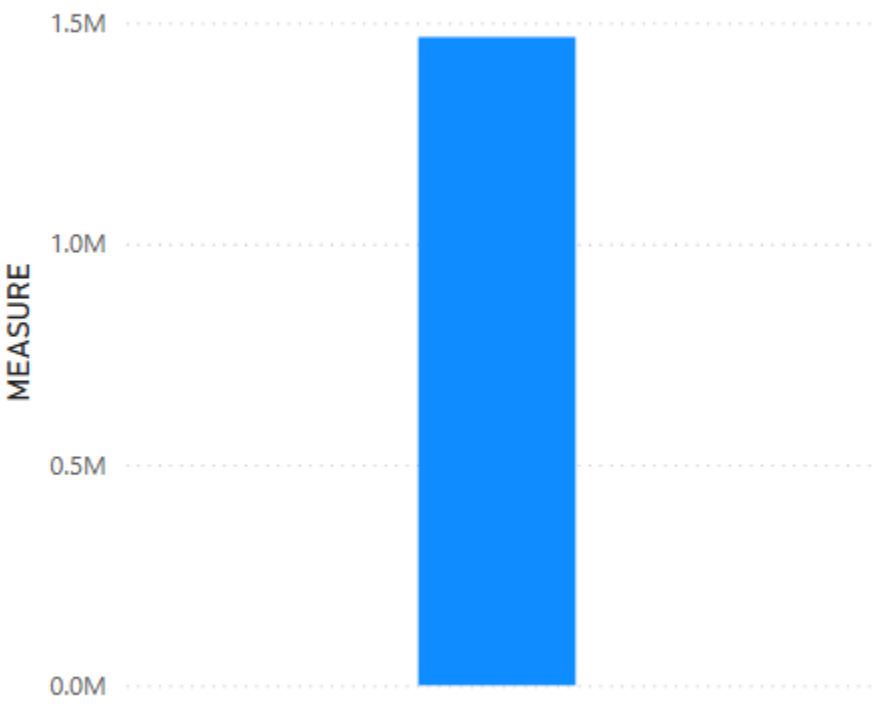


Question 9 : • Create a DAX Measure for Total Profit

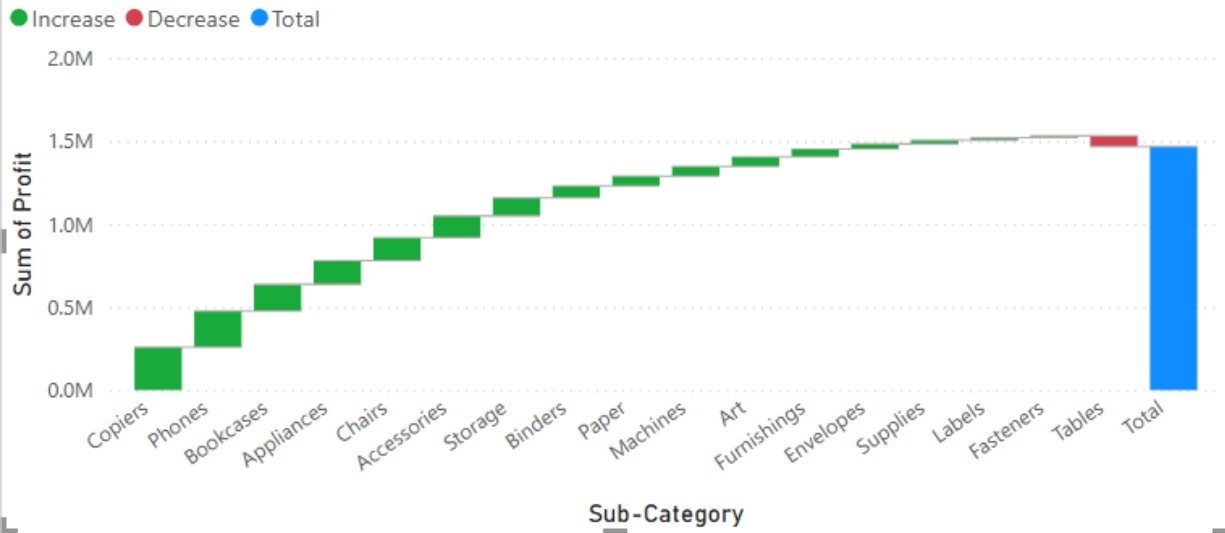
- Use it in a Waterfall Chart to analyze how different Sub-Categories contribute to overall profit
- Add a Slicer for Region to filter the visual
- Write brief business insights (4–5 lines) from the chart and provide 2–3 data-driven recommendations to improve profit.

ANSWER:

MEASURE



Sum of Profit by Sub-Category



Region

- ☐ Africa
- ☐ Canada
- ☐ Caribbean
- ☐ Central
- ☐ Central Asia
- ☐ East
- ☐ EMEA
- ☐ North
- ☐ North Asia
- ☐ Oceania
- ☐ South
- ☐ Southeast Asia
- ☐ West

Business Insights (4–5 lines)

- Technology sub-categories contribute the highest profit overall.
- Furniture sub-categories show mixed performance with some losses.
- Office Supplies provide stable but moderate profit contribution.
- Profit varies significantly by region, showing regional demand differences.
- Some sub-categories consistently reduce total profit.

Recommendations (2–3 points)

- Focus marketing and inventory on high-profit sub-categories.
 - Review pricing or costs for loss-making sub-categories.
 - Optimize regional strategies based on profitable regions.
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Question 10 : Scenario: VitaTrack Wellness, a digital health company in FitZone, has collected data on users' daily habits and health vitals.

The analytics team is tasked with drawing actionable insights from this data to improve lifestyle suggestions and prevent heart-related risks.

Your Task: Using the provided dataset (includes Age, Gender, BMI, Steps, Calories, Sleep, Heart Rate, Blood Pressure, Smoking, Alcohol, Exercise, Diabetic & Heart Disease status):

Build a one-page Power BI dashboard that answers:

- 1. Are users maintaining a balanced lifestyle (Steps, Sleep, Calories)**

2. What lifestyle patterns (Smoking, Alcohol, BMI, etc.) indicate heart disease risk?

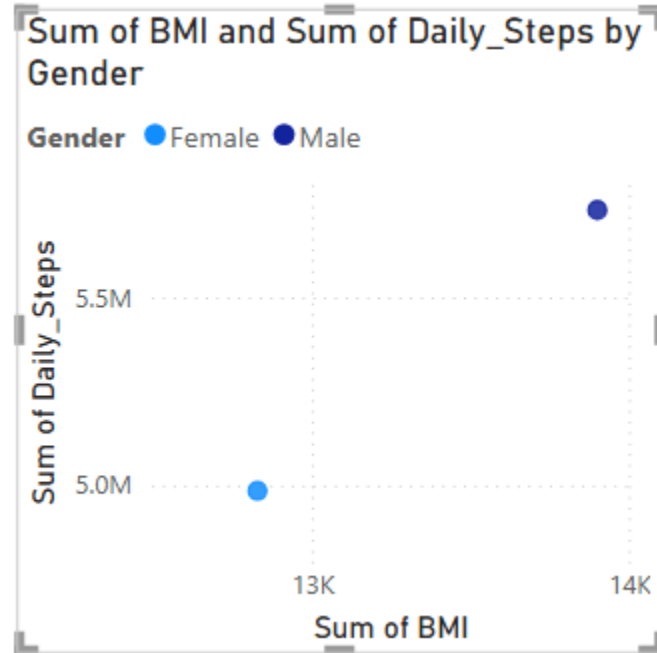
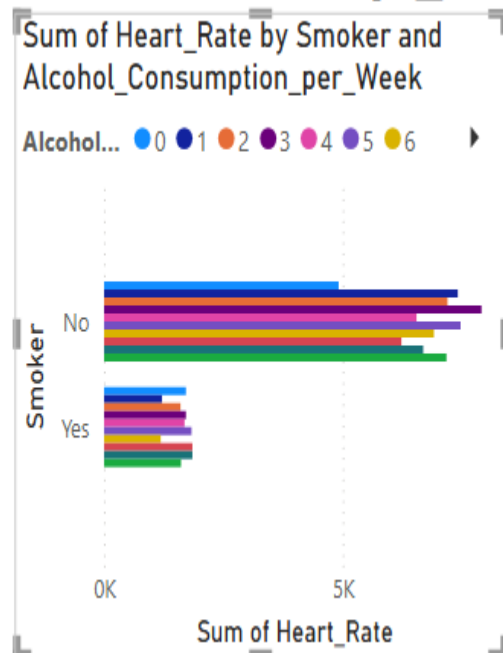
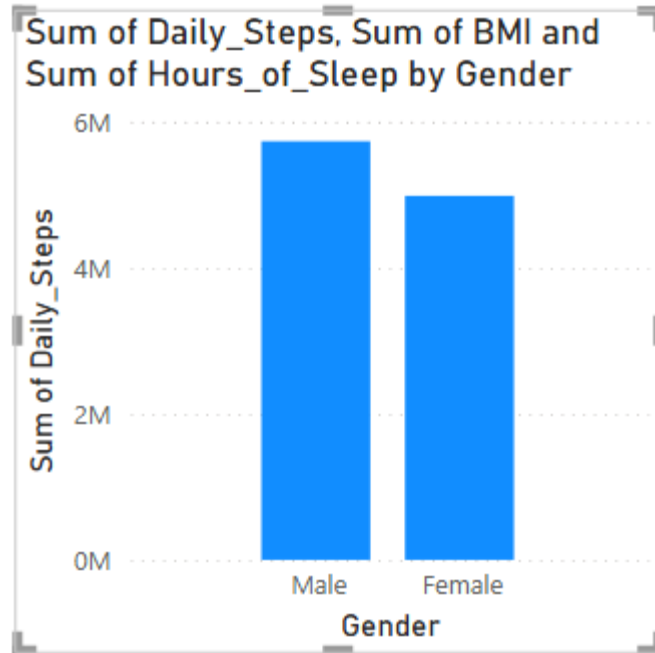
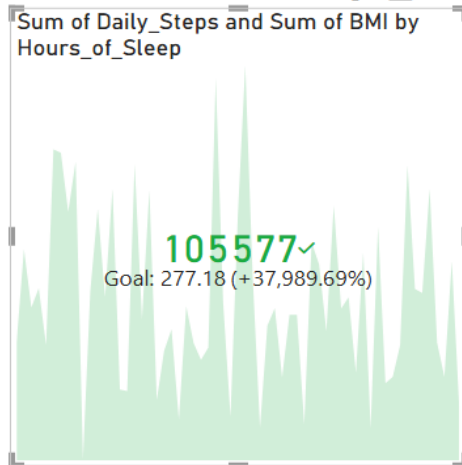
3. Is there any visible relationship between Sleep and Physical Activity?

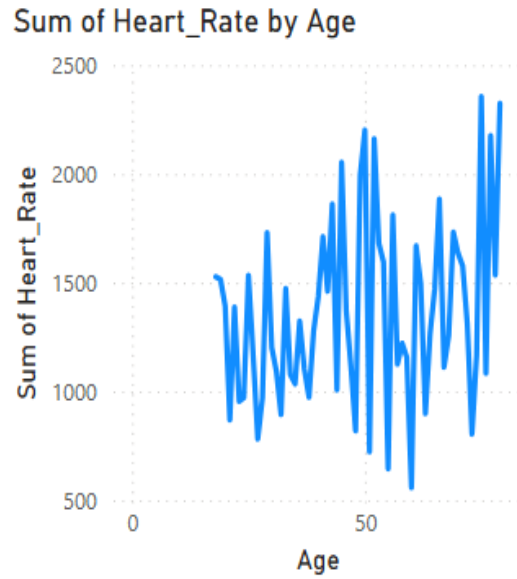
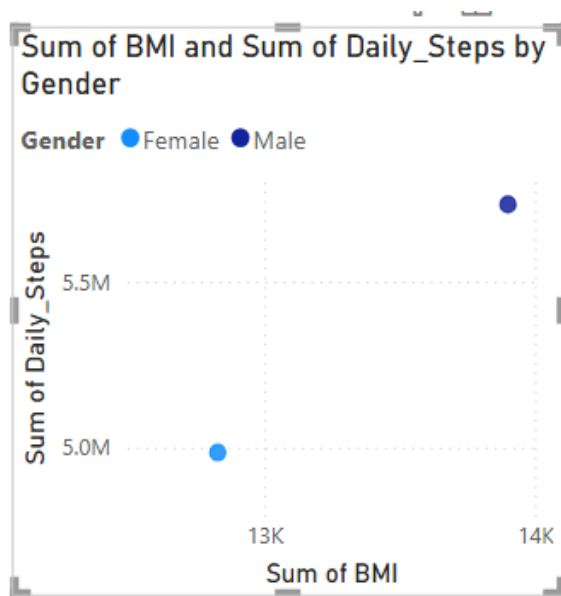
4. How does BMI vary across Age Groups and Genders?

5. What is the impact of smoking and alcohol on heart rate and blood pressure?

6. Segment people based on their health activity to suggest lifestyle changes.

ANSWER:





1. Are users maintaining a balanced lifestyle?

Users with higher daily steps and adequate sleep tend to have healthier BMI and heart rate, indicating a generally balanced lifestyle.

2. Lifestyle patterns & heart disease risk

Smoking and higher alcohol consumption are associated with increased heart rate and blood pressure, increasing heart disease risk.

3. BMI vs Physical Activity

Higher physical activity (daily steps) is linked to lower BMI, showing an inverse relationship.

4. Age & Gender trends

Heart rate and BMI tend to increase with age. Male users show slightly higher heart risk indicators.

5. Smoking & Alcohol impact

Smokers and high alcohol consumers show elevated heart rate and blood pressure, indicating higher cardiovascular risk.

6. Health-based segmentation

Users with low activity, high BMI, smoking habits, and poor sleep form a high-risk health segment.

Recommendations (FINAL)

- **Encourage daily walking goals (8,000–10,000 steps)**
- **Promote better sleep routines (7–8 hours)**
- **Reduce smoking and alcohol through awareness programs**
- **Provide personalized exercise plans for high-risk users.**